frq9

1. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

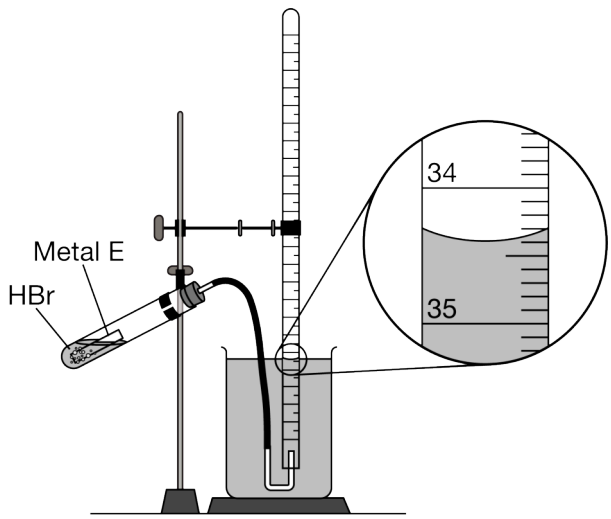
While investigating the chemical and physical properties of metallic element E, a student conducts an experiment by combining $\mathrm{E}(s)$ and $\mathrm{HBr}(aq)$, which react according to the following equation.

$$2\ \mathrm{E}(s) + 2\ \mathrm{HBr}(aq) \rightarrow 2\ \mathrm{EBr}(aq) + \mathrm{H_2}(g)$$

(a) Write the net-ionic equation for the reaction.

(b) In the reaction, is $\mathrm{E}(s)$ being oxidized or reduced? Justify your answer in terms of oxidation numbers.

As shown in the following diagram, $10.00\ \mathrm{mL}$ samples of $\mathrm{HBr}(aq)$ are added to different quantities of $\mathrm{E}(s)$ in a closed test tube. At the start of each trial, the gas-collecting tube is filled completely with distilled water. The samples of $\mathrm{E}(s)$ react with $\mathrm{HBr}(aq)$, and the $\mathrm{H_2}(g)$ produced is collected in the gas-collecting tube.



(c) The enlarged view of the gas-collecting tube at the end of trial 3 is shown in the diagram above. What should the student record as the volume, in mL, of $\mathrm{H_2}(g)$ collected in trial 3?

(d) In the laboratory, the temperature is $295\ \mathrm{K}$ and the total pressure in the gas-collecting tube is $748.6\ \mathrm{mmHg}$. If the vapor pressure of water at $295\ \mathrm{K}$ is $19.8\ \mathrm{mmHg}$, determine the pressure of the $\mathrm{H_2}(g)$ in the gas-collecting tube.

frq9

Trial	Initial Moles of $E(s)$	Initial $[HBr(aq)]$ (M)	Volume of $H_2(g)$ Produced (mL)
1	0.00220	0.500	27.02
2	0.00240	0.500	29.40
3	0.00280	0.500	?

(e) The data from all three trials are given above. A student claims that $HBr(aq)$ is the limiting reactant in trial 1. Do you agree or disagree? Justify your answer with information in the data table.

(f) A certain pure compound containing element E is analyzed. The elemental composition of the compound is shown in the following table. Determine the empirical formula of the compound using the information in the table.

Element	Moles Present in the Compound
E	0.0676
S	0.0338
O	0.102

(g) A different element, L, is located in the same row of the periodic table as element E, but L has roughly half the atomic mass of E. Is the atomic radius of L less than, equal to, or greater than the atomic radius of E? Justify your answer using principles of atomic structure.

(h) The ionic radii of elements E and a different metallic element, M, are shown in the following table. Both elements form oxides, E_2O and MO. If lattice energy is defined as the energy required to separate an ionic solid into individual separate gaseous ions, would the lattice energy of MO be less than, equal to, or greater than the lattice energy of the oxide E_2O? Justify your answer in terms of Coulomb's law.

Element	Ionic radius (pm)
E	170
M	110

frq9

2. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

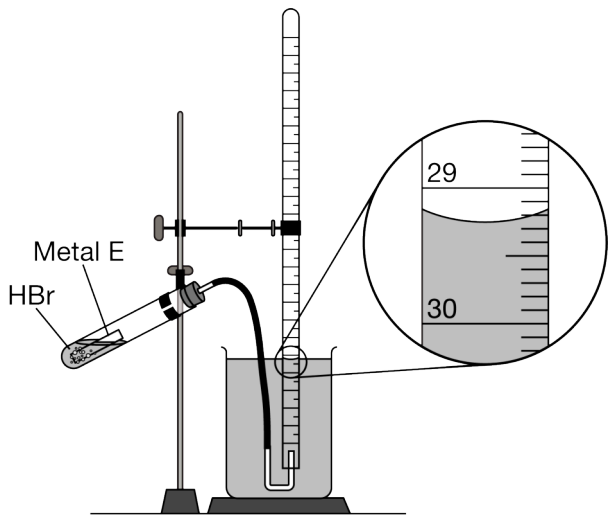
While investigating the chemical and physical properties of metallic element E, a student conducts an experiment by combining $\mathrm{E}(s)$ and $\mathrm{HBr}(aq)$, which react according to the following equation.

$$2\ \mathrm{E}(s) + 6\ \mathrm{HBr}(aq) \rightarrow 2\ \mathrm{EBr_3}(aq) + 3\mathrm{H_2}(g)$$

(a) Write the net-ionic equation for the reaction.

(b) In the reaction, is $\mathrm{E}(s)$ being oxidized or reduced? Justify your answer in terms of oxidation numbers.

As shown in the following diagram, $10.00\ \mathrm{mL}$ samples of $\mathrm{HBr}(aq)$ are added to different quantities of $\mathrm{E}(s)$ in a closed test tube. At the start of each trial, the gas-collecting tube is filled completely with distilled water. The samples of $\mathrm{E}(s)$ react with $\mathrm{HBr}(aq)$, and the $\mathrm{H_2}(g)$ produced is collected in the gas-collecting tube.



(c) The enlarged view of the gas-collecting tube at the end of trial 3 is shown in the diagram above. What should the student record as the volume, in mL, of $\mathrm{H_2}(g)$ collected in trial 3?

(d) In the laboratory, the temperature is $295\ \mathrm{K}$ and the total pressure in the gas-collecting tube is $744.7\ \mathrm{mmHg}$. If the vapor pressure of water at $295\ \mathrm{K}$ is $19.8\ \mathrm{mmHg}$, determine the pressure of the $\mathrm{H_2}(g)$ in the gas-collecting tube.

frq9

Trial	Initial Moles of $E(s)$	Initial $[HBr(aq)]$ (M)	Volume of $H_2(g)$ Produced (mL)
1	5.40×10^{-4}	0.400	19.95
2	6.90×10^{-4}	0.400	25.52
3	8.20×10^{-4}	0.400	?

(e) The data from all three trials are given above. A student claims that $E(s)$ is the limiting reactant in trial 2. Do you agree or disagree? Justify your answer with information in the data table.

(f) A certain pure compound containing element E is analyzed. The elemental composition of the compound is shown in the following table. Determine the empirical formula of the compound using the information in the table.

Element	Moles Present in the Compound
E	0.0167
I	0.050
O	0.200

(g) A different element, L, is located in the same column of the periodic table as element E, but L has roughly double the atomic mass of E. Is the atomic radius of L less than, equal to, or greater than the atomic radius of E? Justify your answer using principles of atomic structure.

(h) The ionic radii of ions X^{2-} and Z^{2-} are shown in the following table. Element E can bond with X or Z to form E_2X_3 or E_2Z_3, respectively. If lattice energy is defined as the energy required to separate an ionic solid into individual separate gaseous ions, would the lattice energy of E_2X_3 be less than, equal to, or greater than the lattice energy of E_2Z_3? Justify your answer in terms of Coulomb's law.

Element	Ionic radius (pm)
X^{2-}	120
Z^{2-}	180

frq9

3. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student wishes to determine the concentration of $Ag^+(aq)$ in a solution of $AgNO_3(aq)$. The student combines $10.00\ mL$ of $AgNO_3(aq)$ with excess $Na_2SO_4(aq)$ and observes the formation of a white precipitate. The formation of the precipitate is represented by the following equation.

$$2\ AgNO_3(aq) + Na_2SO_4(aq) \rightarrow Ag_2SO_4(s) + 2\ NaNO_3(aq)$$

(a) Write the balanced net ionic equation for the precipitation reaction.

The student collects the precipitate by filtration and measures the mass of the filter paper and precipitate every 10 minutes as it dries. The student records the data in the following table.

Mass of dry filter paper	$0.88\ g$
Mass of filter paper and precipitate immediately after filtration	$4.82\ g$
Mass of filter paper and precipitate after 10 minutes	$4.37\ g$
Mass of filter paper and precipitate after 20 minutes	$4.01\ g$
Mass of filter paper and precipitate after 30 minutes	$3.79\ g$

(b) Use the data above to calculate the number of moles of $Ag_2SO_4(s)$ (molar mass $311.8\ g/mol$) that precipitated.

(c) Calculate the concentration of $Ag^+(aq)$ in the original $10.00\ mL$ solution of $AgNO_3(aq)$.

(d) The concentration of $Ag^+(aq)$ determined by the student is significantly higher than the actual concentration of $Ag^+(aq)$. Based on the student's data table, identify an error in the experimental procedure that led to this result.

frq9

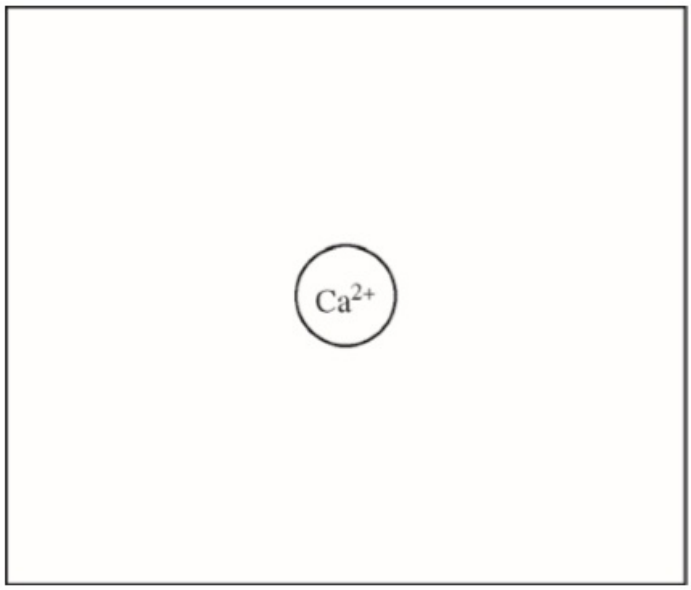
4. Answer the following questions about the solubility of $Ca(OH)_2$ ($K_{sp} = 1.3 \times 10^{-6}$) .

a. Write a balanced chemical equation for the dissolution of $Ca(OH)_2(s)$ in pure water.

b. Calculate the molar solubility of $Ca(OH)_2$ in 0.10 *M* $Ca(NO_3)_2$.

c. In the box below, complete a particle representation diagram that includes four water molecules with proper
orientation around the Ca^{2+} ion.

Represent water molecules as .

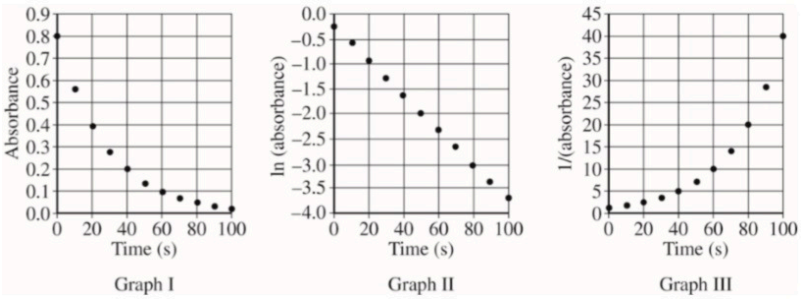


frq9

5.

$$Na_2C_{37}H_{34}N_2S_3O_9 + OCl^- \rightarrow \text{products}$$
blue colorless

Blue food coloring can be oxidized by household bleach (which contains OCl^-) to form colorless products, as represented by the equation above. A student used a spectrophotometer set at a wavelength of 635 nm to study the absorbance of the food coloring over time during the bleaching process. In the study, bleach is present in large excess so that the concentration of OCl^- is essentially constant throughout the reaction. The student used data from the study to generate the graphs below.



a. Based on the graphs above, what is the order of the reaction with respect to the blue food coloring?

b. The reaction is known to be first order with respect to bleach. In a second experiment, the student prepares solutions of food coloring and bleach with concentrations that differ from those used in the first experiment. When the solutions are combined, the student observes that the reaction mixture reaches an absorbance near zero too rapidly. In order to correct the problem, the student proposes the following three possible modifications to the experiment.

• Increasing the temperature
• Increasing the concentration of the food coloring
• Increasing the concentration of the bleach

Circle the one proposed modification above that could correct the problem, and explain how that modification increases the time for the reaction mixture to reach an absorbance near zero.

c. In another experiment, a student wishes to study the oxidation of red food coloring with bleach. How would the student need to modify the original experimental procedure to determine the order of the reaction with respect to the red food coloring?

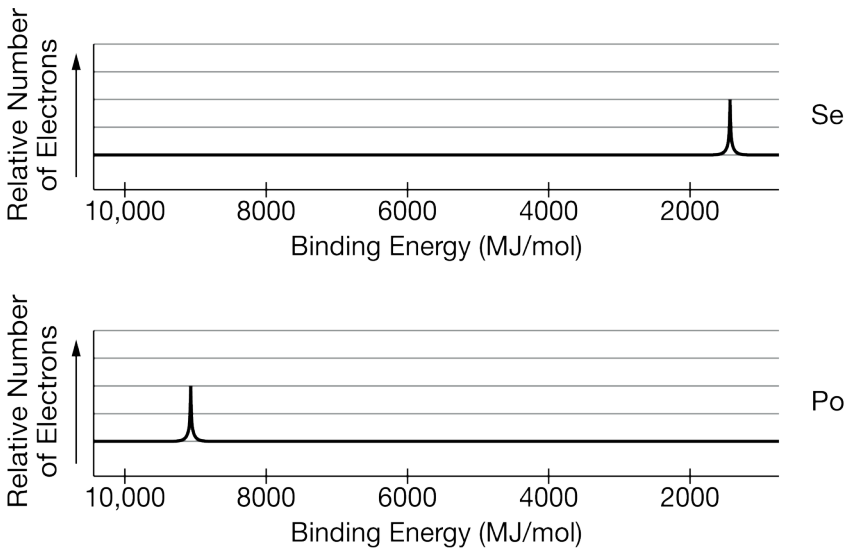
frq9

6. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

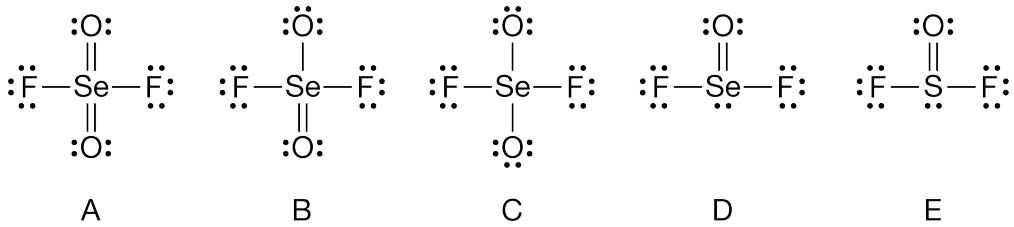
Answer the following questions about elements in Group 16 of the periodic table.

(a) Using principles of atomic structure, explain why the first ionization energy of Po is less than that of Se.

(b) Photoelectron spectroscopy data for the $1s$ sublevel of Se and the $1s$ sublevel of Po are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for Po is positioned so far to the left of the peak for Se.



Three compounds that contain elements from Group 16 are $\mathrm{SeO_2F_2}$, $\mathrm{SeOF_2}$, and $\mathrm{SOF_2}$. Possible Lewis electron dot diagrams for the compounds are shown below.



(c) Which structure, A, B, or C, minimizes the formal charges for each atom in $\mathrm{SeO_2F_2}$? Justify your choice.

(d) A student proposes that the energy of the $\mathrm{Se-to-O}$ bond in the structure in diagram D is greater than that of the $\mathrm{Se-to-O}$ bond in the structure in diagram C. Do you agree or disagree with the student's claim? Justify your answer.

(e) The dipole moment of the $\mathrm{SeO_2F_2}$ molecule (diagrams A, B and C) is nonzero. Identify its molecular geometry and explain how its geometry and the polarity of its bonds result in a nonzero dipole moment.

frq9

The melting point of $SeOF_2$ (diagram D) is greater than the melting point of SOF_2 (diagram E).

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in melting points based on the types and relative strengths of intermolecular forces.

(h) $SeOF_2$ reacts with SiO_2 to produce SeO_2 and SiF_4. The unbalanced chemical equation for the reaction appears below.

$$a\ SeOF_2 + b\ SiO_2 \rightarrow c\ SeO_2 + d\ SiF_4$$

Give the value of each of the following coefficients in the balanced equation.

$a =$
$b =$
$c =$
$d =$

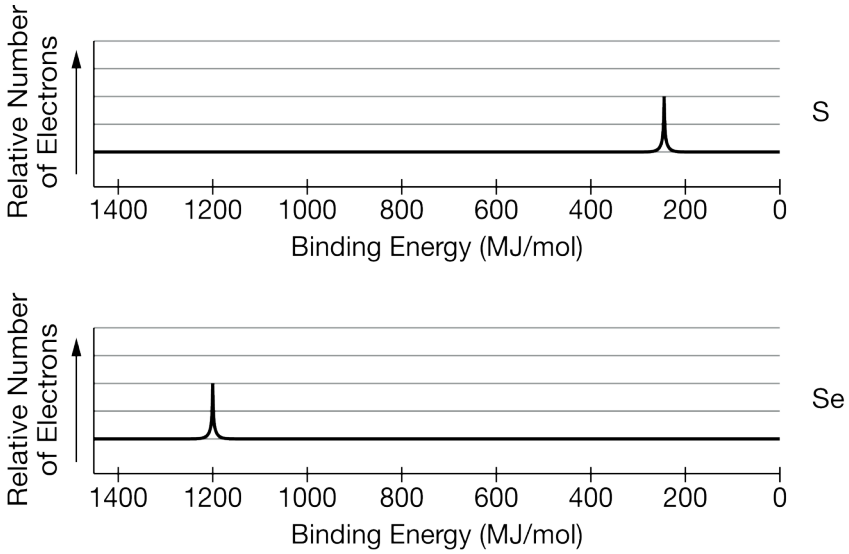
frq9

7. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

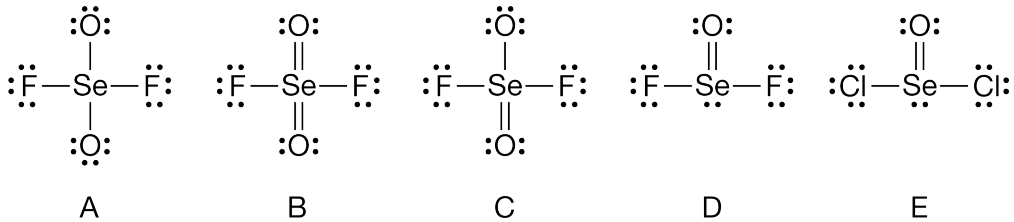
Answer the following questions about elements in Group 16 of the periodic table.

(a) Using principles of atomic structure, explain why the atomic radius of Kr is less than the ionic radius of Se^{2-}.

(b) Photoelectron spectroscopy data for the $1s$ sublevel of S and the $1s$ sublevel of Se are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for Se is positioned so far to the left of the peak for S.



Three compounds that contain elements from Group 16 are $\mathrm{SeO_2F_2}$, $\mathrm{SeOF_2}$, and $\mathrm{SeOCl_2}$. Possible Lewis electron dot diagrams for the compounds are shown below.



(c) Which structure, A, B, or C, minimizes the formal charges for each atom in $\mathrm{SeO_2F_2}$? Justify your choice.

(d) A student proposes that the energy of the $\mathrm{Se-to-O}$ bond in the structure in diagram D is less than that of the $\mathrm{Se-to-O}$ bond in the structure in diagram A. Do you agree or disagree with the student's claim? Justify your answer.

frq9

(e) The dipole moment of the $SeOF_2$ molecule (diagram D) is nonzero. Identify its molecular geometry and explain how its geometry and the polarity of its bonds result in a nonzero dipole moment.

The boiling point of $SeOF_2$ (diagram D) is less than the boiling point of $SeOCl_2$ (diagram E).

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in boiling points based on the types and relative strengths of intermolecular forces.

(h) $SeOF_2$ reacts with $NaOH$ to produce Na_2SeO_3, NaF, and H_2O. The unbalanced chemical equation for the reaction appears below.

$$a\ SeOF_2 + b\ NaOH \rightarrow c\ Na_2SeO_3 + d\ NaF + e\ H_2O$$

Give the value of each of the following coefficients in the balanced equation.

$a =$
$b =$
$c =$
$d =$
$e =$

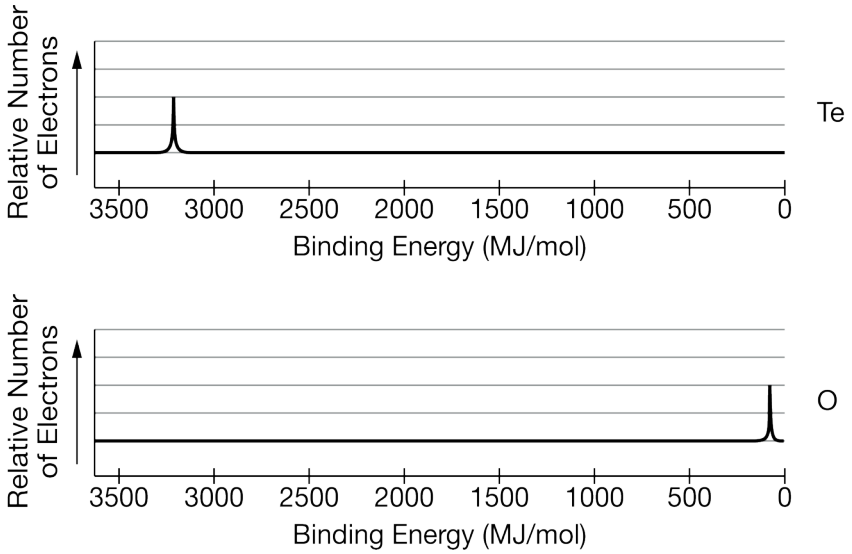
frq9

8. **Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate**.

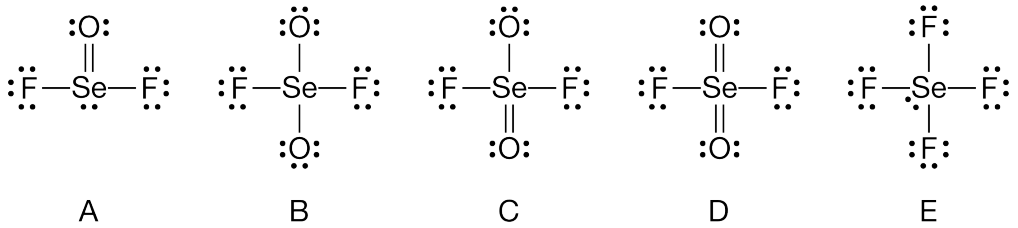
Answer the following questions about elements in Group 16 of the periodic table.

(a) Using principles of atomic structure, explain why the atomic radius of Te is less than the ionic radius of Te^{2-}.

(b) Photoelectron spectroscopy data for the $1s$ sublevel of Te and the $1s$ sublevel of O are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for O is positioned so far to the right of the peak for Te.



Three compounds that contain elements from Group 16 are $\mathrm{SeOF_2}$, $\mathrm{SeO_2F_2}$, and $\mathrm{SeF_4}$. Possible Lewis electron dot diagrams for the compounds are shown below.



(c) Which structure, B, C or D, minimizes the formal charges for each atom in $\mathrm{SeO_2F_2}$? Justify your choice.

(d) A student proposes that the energy of the $\mathrm{Se-to-F}$ bond in the structure in diagram A is equal to that of the $\mathrm{Se-to-F}$ bond in the structure in diagram D. Do you agree or disagree with the student's claim? Justify your answer.

frq9

(e) The dipole moment of the SeF_4 molecule (diagram E) is nonzero. Identify its molecular geometry and explain how its geometry and the polarity of its bonds result in a nonzero dipole moment.

The melting point of $SeOF_2$ (diagram A) is greater than the melting point of SeO_2F_2 (diagrams B, C, and D).

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in melting points based on the types and relative strengths of intermolecular forces.

(h) TeO_2 reacts with SeF_4 to produce $SeOF_2$ and TeF_4. The unbalanced chemical equation for the reaction appears below.

$a\ TeO_2 + b\ SeF_4 \rightarrow c\ SeOF_2 + d\ TeF_4$

Give the value of each of the following coefficients in the balanced equation.

$a =$
$b =$
$c =$
$d =$

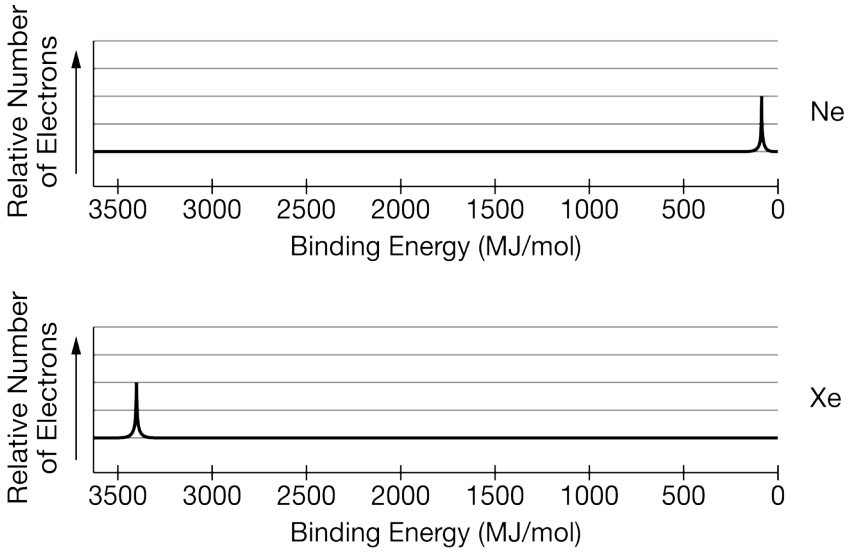
frq9

9. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

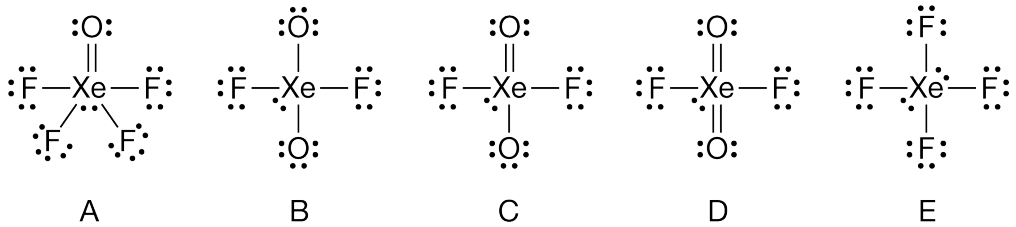
Answer the following questions about elements in Group 18 of the periodic table.

(a) Using principles of atomic structure, explain why the atomic radius of Xe is less than that of Te.

(b) Photoelectron spectroscopy data for the $1s$ sublevel of Ne and the $1s$ sublevel of Xe are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for Ne is positioned so far to the right of the peak for Xe.



Three compounds that contain an element from Group 18 are $\mathrm{XeOF_4}$, $\mathrm{XeO_2F_2}$, and $\mathrm{XeF_4}$. Possible Lewis electron dot diagrams for the compounds are shown below.



(c) Which structure, B, C, or D, minimizes the formal charges for each atom in $\mathrm{XeO_2F_2}$? Justify your choice.

(d) A student proposes that the energy of the $\mathrm{Xe-to-F}$ bond in the structure in diagram B is equal to that of the $\mathrm{Xe-to-F}$ bond in the structure in diagram A. Do you agree or disagree with the student's claim? Justify your answer.

(e) The dipole moment of the $\mathrm{XeOF_4}$ molecule (diagram A) is nonzero. Identify its molecular geometry and

frq9

explain how its geometry and the polarity of its bonds result in a nonzero dipole moment.

The boiling point of XeF_4 (diagram E) is greater than the boiling point of elemental Xe.

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in boiling points based on the types and relative strengths of intermolecular forces.

(h) $XeOF_4$ reacts with SiO_2 to produce XeO_2F_2 and SiF_4. The unbalanced chemical equation for the reaction appears below.

$$a\ SiO_2 + b\ XeOF_4 \rightarrow c\ SiF_4 + d\ XeO_2F_2$$

Give the value of each of the following coefficients in the balanced equation.

$a =$
$b =$
$c =$
$d =$

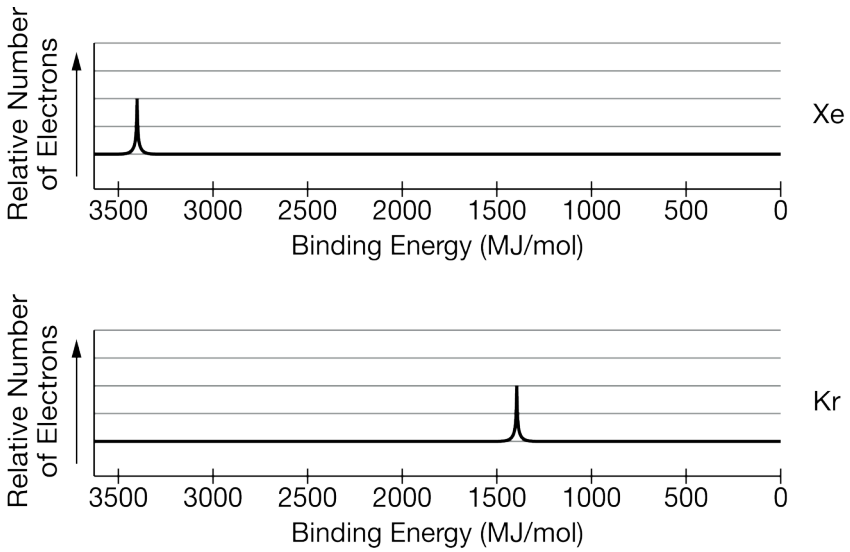
frq9

10. **Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

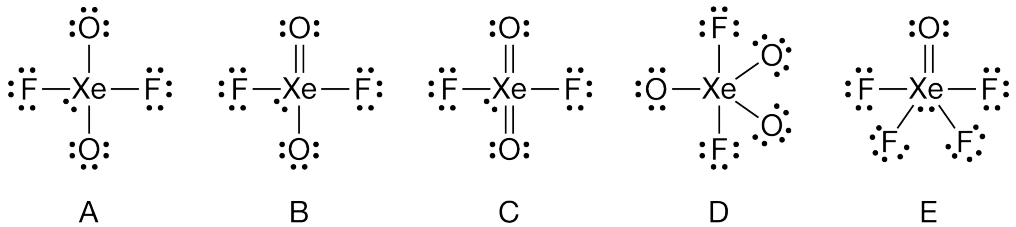
Answer the following questions about elements in Group 18 of the periodic table.

(a) Using principles of atomic structure, explain why the ionic radius of Cs^+ is less than the atomic radius of Xe.

(b) Photoelectron spectroscopy data for the $1s$ sublevel of Xe and the $1s$ sublevel of Kr are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for Kr is positioned so far to the right of the peak for Xe.



Three compounds that contain an element from Group 18 are XeO_2F_2, XeO_3F_2, and $XeOF_4$. Possible Lewis electron dot diagrams for the compounds are shown below.



(c) Which structure, A, B, or C, minimizes the formal charges for each atom in XeO_2F_2? Justify your choice.

(d) A student proposes that the energy of the $Xe - to - O$ bond in the structure in diagram E is greater than that of the $Xe - to - O$ bond in the structure in diagram D. Do you agree or disagree with the student's claim? Justify your answer.

(e) The dipole moment of the XeO_3F_2 molecule (diagram D) is zero. Identify its molecular geometry and explain how its geometry and the polarity of its bonds result in a zero dipole moment.

frq9

The melting point of $XeOF_4$ (diagram E) is greater than the melting point of XeO_3F_2 (diagram D).

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in melting points based on the types and relative strengths of intermolecular forces.

(h) XeF_6 reacts with SiO_2 to produce $XeOF_4$ and SiF_4. The unbalanced chemical equation for the reaction appears below.

$$a\ XeF_6 + b\ SiO_2 \rightarrow c\ XeOF_4 + d\ SiF_4$$

Give the value of each of the following coefficients in the balanced equation.

$a =$
$b =$
$c =$
$d =$

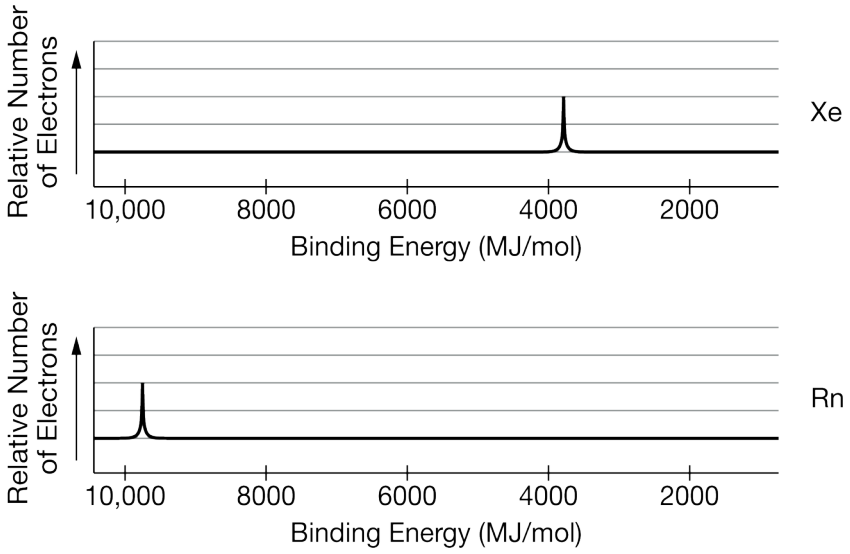
frq9

11. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

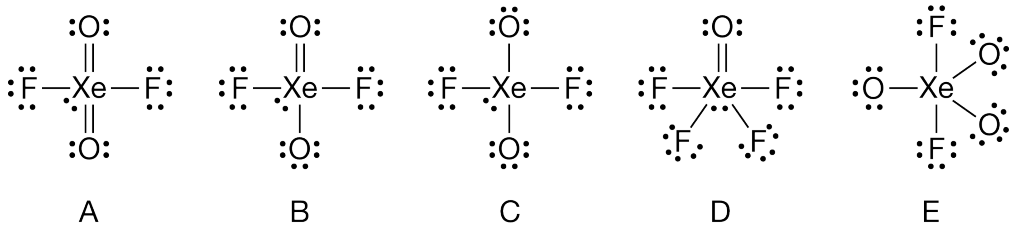
Answer the following questions about elements in Group 18 of the periodic table.

(a) Using principles of atomic structure, explain why the first ionization energy of Rn is less than that of Xe.

(b) Photoelectron spectroscopy data for the $1s$ sublevel of Xe and the $1s$ sublevel of Rn are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for Xe is positioned so far to the right of the peak for Rn.



Three compounds that contain an element from Group 18 are $\mathrm{XeO_2F_2}$, $\mathrm{XeOF_4}$, and $\mathrm{XeO_3F_2}$. Possible Lewis electron-dot diagrams for the compounds are shown below.



(c) Which structure, A, B or C, minimizes the formal charges for each atom in $\mathrm{XeO_2F_2}$? Justify your choice.

(d) A student proposes that the energy of the $\mathrm{Xe-to-O}$ bond in the structure in diagram D is less than that of the $\mathrm{Xe-to-O}$ bond in the structure in diagram E. Do you agree or disagree with the student's claim? Justify your answer.

(e) The dipole moment of the $\mathrm{XeO_2F_2}$ molecule (diagrams A, B and C) is nonzero. Identify its molecular

frq9

geometry and explain how its geometry and the polarity of its bonds result in a nonzero dipole moment.

The melting point of XeO_2F_2 (diagrams A, B, C) is greater than the melting point of XeO_3F_2 (diagram E).

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in melting points based on the types and relative strengths of intermolecular forces.

(h) XeO_2F_2 reacts with SiO_2 to produce SiF_4 and XeO_3. The unbalanced chemical equation for the reaction appears below.

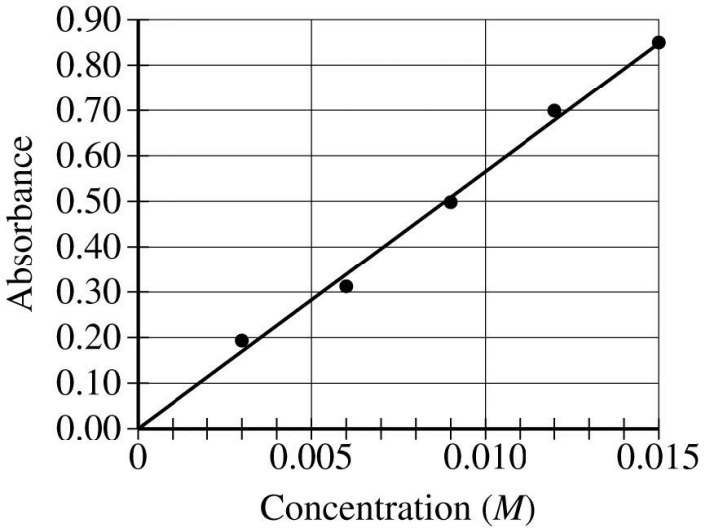
$$a\ XeO_2F_2 + b\ SiO_2 \rightarrow c\ SiF_4 + d\ XeO_3$$

Give the value of each of the following coefficients in the balanced equation.

$a =$
$b =$
$c =$
$d =$

frq9

12. To spectrophotometrically determine the mass percent of cobalt in an ore containing cobalt and some inert materials, solutions with known $[Co^{2+}]$ are prepared and the absorbance of each of the solutions is measured at the wavelength of optimum absorbance. The data are used to create a calibration plot, shown below.



A 0.630 g sample of the ore is completely dissolved in concentrated $HNO_3(aq)$. The mixture is diluted with water to a final volume of 50.00 mL. Assume that all the cobalt in the ore sample is converted to $Co^{2+}(aq)$.

a. What is the $[Co^{2+}]$ in the solution if the absorbance of a sample of the solution is 0.74?

b. Calculate the number of moles of $Co^{2+}(aq)$ in the 50.00 mL solution.

c. Calculate the mass percent of Co in the 0.630 g sample of the ore.

frq9

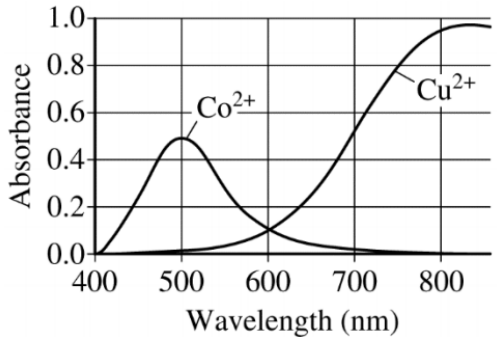
13. **Directions:** Examples and equations may be included in your responses where appropriate. For calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Pay attention to significant figures.

A student has 100. mL of 0.400 M $CuSO_4(aq)$ and is asked to make 100. mL of 0.150 M $CuSO_4(aq)$ for a spectrophotometry experiment. The following laboratory equipment is available for preparing the solution: centigram balance, weighing paper, funnel, 10 mL beaker, 150 mL beaker, 50 mL graduated cylinder, 100 mL volumetric flask, 50 mL buret, and distilled water.

(a) Calculate the volume of 0.400 M $CuSO_4(aq)$ required for the preparation.

(b) Briefly describe the essential steps to most accurately prepare the 0.150 M $CuSO_4(aq)$ from the 0.400 M $CuSO_4(aq)$ using the equipment listed above.

The student plans to conduct a spectrophotometric analysis to determine the concentration of $Cu^{2+}(aq)$ in a solution. The solution has a small amount of $Co(NO_3)_2(aq)$ present as a contaminant. The student is given the diagram below, which shows the absorbance curves for aqueous solutions of $Co^{2+}(aq)$ and $Cu^{2+}(aq)$.



(c) The spectrophotometer available to the student has a wavelength range of 400 nm to 700 nm. What wavelength should the student use to minimize the interference from the presence of the $Co^{2+}(aq)$ ions?

frq9

14. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

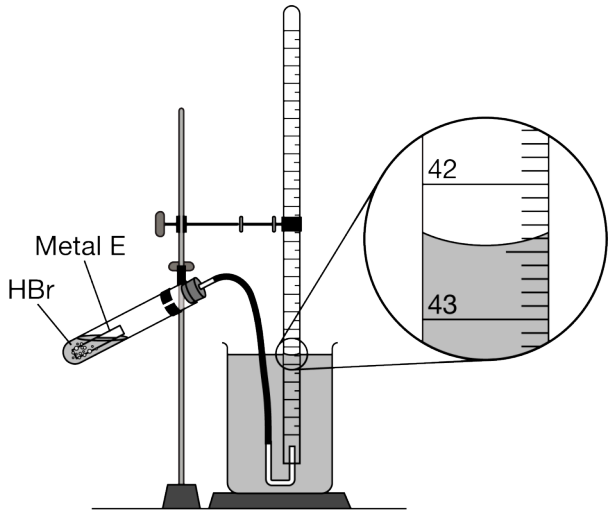
While investigating the chemical and physical properties of metallic element E, a student conducts an experiment by combining $\mathrm{E}(s)$ and $\mathrm{HBr}(aq)$, which react according to the following equation.

$$\mathrm{E}(s) + 2\,\mathrm{HBr}(aq) \rightarrow \mathrm{EBr_2}(aq) + \mathrm{H_2}(g)$$

(a) Write the net-ionic equation for the reaction.

(b) In the reaction, is $\mathrm{E}(s)$ oxidized or reduced? Justify your answer in terms of oxidation numbers.

As shown in the following diagram, $10.00\,\mathrm{mL}$ samples of $\mathrm{HBr}(aq)$ of various concentrations are added to samples of $\mathrm{E}(s)$ in a closed test tube. At the start of each trial, the gas-collecting tube is filled completely with distilled water. The samples of $\mathrm{E}(s)$ react with $\mathrm{HBr}(aq)$, and the $\mathrm{H_2}(g)$ produced is collected in the gas-collecting tube.



(c) The enlarged view of the gas-collecting tube at the end of trial 3 is shown in the diagram above. What should the student record as the volume, in mL, of $\mathrm{H_2}(g)$ collected in trial 3?

(d) In the laboratory, the temperature is $294\,\mathrm{K}$ and the total pressure in the gas-collecting tube is $754.7\,\mathrm{mmHg}$. If the vapor pressure of water at $294\,\mathrm{K}$ is $18.6\,\mathrm{mmHg}$, determine the pressure of the $\mathrm{H_2}(g)$ in the gas-collecting tube.

frq9

Trial	Initial Moles of $E(s)$	Initial $[HBr(aq)]$ (M)	Volume of $H_2(g)$ Produced (mL)
1	0.00150	0.200	24.52
2	0.00150	0.250	30.95
3	0.00150	0.350	?

(e) The data from all three trials are given above. A student claims that $E(s)$ is the limiting reactant in trial 1. Do you agree or disagree? Justify your answer with information in the data table.

(f) A certain pure compound containing element E is analyzed. The elemental composition of the compound is shown in the following table. Determine the empirical formula of the compound using the information in the table.

Element	Moles Present in the Compound
E	0.0652
C	0.130
O	0.260

(g) A different element, L, is located in the same row of the periodic table as element E, but L has roughly double the atomic mass of E. Is the atomic radius of L less than, equal to, or greater than the atomic radius of E? Justify your answer using principles of atomic structure.

(h) The ionic radii of elements E and a different metallic element, M, are shown in the following table. Both elements form oxides, EO and M_2O. If lattice energy is defined as the energy required to separate an ionic solid into individual separate gaseous ions, would the lattice energy of M_2O be less than, equal to, or greater than the lattice energy of the oxide EO? Justify your answer in terms of Coulomb's law.

Element	Ionic radius (pm)
E	110
M	150

frq9

15. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

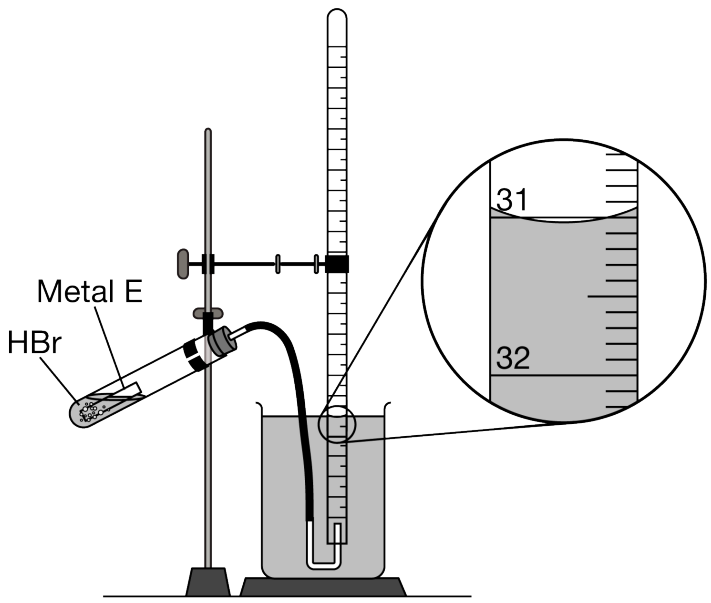
While investigating the chemical and physical properties of metallic element E, a student conducts an experiment by combining $\mathrm{E}(s)$ and $\mathrm{HBr}(aq)$, which react according to the following equation.

$$\mathrm{E}(s) + 4\,\mathrm{HBr}(aq) \rightarrow \mathrm{EBr_4}(aq) + 2\,\mathrm{H_2}(g)$$

(a) Write the net-ionic equation for the reaction.

(b) In the reaction, is H being oxidized or reduced? Justify your answer using oxidation numbers.

As shown in the following diagram, in a closed test tube, $10.00\,\mathrm{mL}$ samples of $\mathrm{HBr}(aq)$ of various concentrations are added to samples of $\mathrm{E}(s)$. At the start of each trial, the gas-collecting tube is filled completely with distilled water. The samples of $\mathrm{E}(s)$ react with $\mathrm{HBr}(aq)$, and the $\mathrm{H_2}(g)$ produced is collected in the gas-collecting tube.



(c) The enlarged view of the gas-collecting tube at the end of trial 3 is shown in the diagram above. What should the student record as the volume, in mL, of $\mathrm{H_2}(g)$ collected in trial 3?

(d) In the laboratory, the temperature is $295\,\mathrm{K}$ and the total pressure in the gas-collecting tube is $741.2\,\mathrm{mmHg}$. If the vapor pressure of water at $295\,\mathrm{K}$ is $19.8\,\mathrm{mmHg}$, determine the pressure of the $\mathrm{H_2}(g)$ in the gas-collecting tube.

frq9

Trial	Initial Moles of $E(s)$	Initial $[HBr(aq)]$ (M)	Volume of $H_2(g)$ Produced (mL)
1	7.50×10^{-4}	0.150	18.60
2	7.50×10^{-4}	0.200	24.81
3	7.50×10^{-4}	0.250	?

(e) The data from all three trials are given above. A student claims that $E(s)$ is the limiting reactant in trial 2. Do you agree or disagree? Justify your answer with information in the data table.

(f) A certain pure compound containing element E is analyzed. The elemental composition of the compound is shown in the following table. Determine the empirical formula of the compound using the information in the table.

Element	Moles Present in the Compound
E	0.0339
C	0.136
O	0.272

(g) A different element, L, is located in the same column of the periodic table as element E, but L has roughly half the atomic mass of E. Is the atomic radius of L less than, equal to, or greater than the atomic radius of E? Justify your answer using principles of atomic structure.

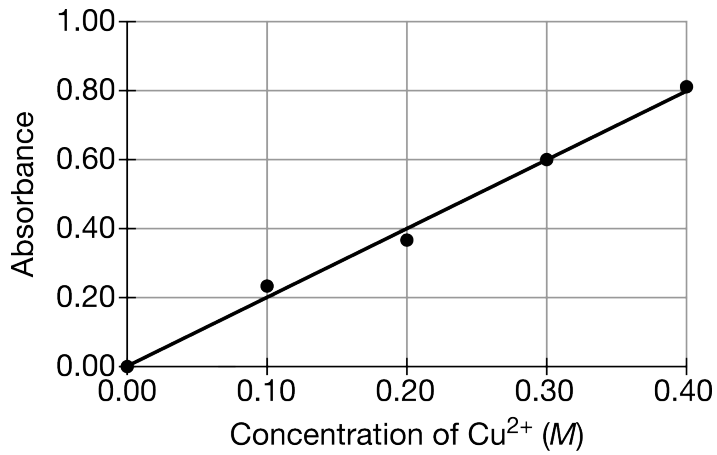
(h) The ionic radii of ions X^- and Z^- are shown in the following table. Element E can bond with X or Z to form EX_4 or EZ_4, respectively. If lattice energy is defined as the energy required to separate an ionic solid into individual separate gaseous ions, would the lattice energy of EX_4 be less than, equal to, or greater than the lattice energy of EZ_4? Justify your answer in terms of Coulomb's law.

Element	Ionic radius (pm)
X^-	170
Z^-	120

frq9

16. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student is given a sample of $CuSO_4(s)$ that contains a solid impurity that is soluble and colorless. The student wants to determine the amount of $CuSO_4$ in the sample and decides to use a spectrophotometer. First, the student prepares a calibration graph by measuring the absorbances of $CuSO_4(aq)$ solutions of known concentrations. The graph is shown below.

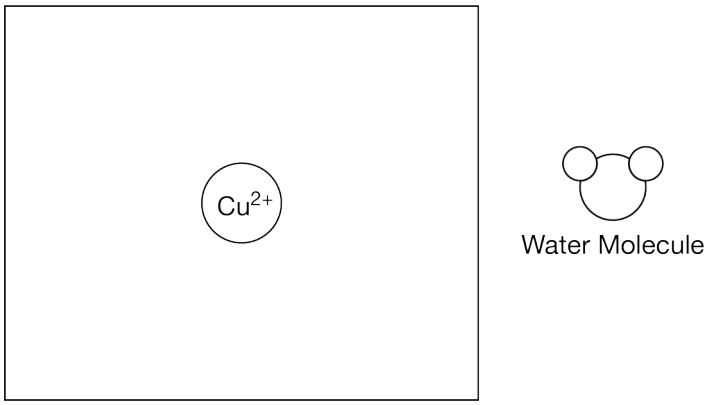


(a) The student dissolves the entire impure sample of $CuSO_4(s)$ in enough distilled water to make $100.\ mL$ of solution. Then the student measures the absorbance of the solution and observes that it is 0.30. Determine the concentration of $CuSO_4(aq)$ in the solution.

(b) Calculate the number of moles of $CuSO_4$ that were in the impure sample of $CuSO_4(s)$.

(c) In addition to the number of moles of $CuSO_4$ calculated in part (b), what other quantity must be measured in order to calculate the mass percentage of $CuSO_4$ in the impure sample of $CuSO_4(s)$?

(d) A particulate drawing of a single Cu^{2+} ion is shown in the following box. Draw <u>two</u> H_2O molecules in the box to show the most likely orientation of $H_2O(l)$ molecules around the Cu^{2+} ion.



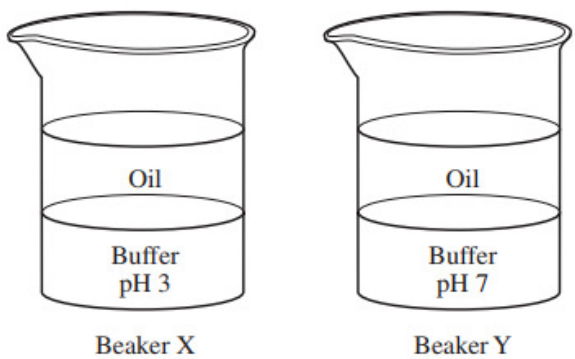
frq9

17.

$$\mathrm{HIn}(aq) + \mathrm{H_2O}(l) \rightleftharpoons \mathrm{In^-}(aq) + \mathrm{H_3O^+}(aq)$$

yellow *blue*

The indicator Hln is a weak acid with a pK_a value of 5.0. It reacts with water as represented in the equation above. Consider the two beakers below. Each beaker has a layer of colorless oil (a nonpolar solvent) on top of a layer of aqueous buffer solution. In beaker X the pH of the buffer solution is 3, and in beaker Y the pH of the buffer solution is 7. A small amount of Hln is placed in both beakers. The mixtures are stirred well, and the oil and water layers are allowed to separate.



a. What is the predominant form of Hln in the aqueous buffer in beaker Y, the acid form or the conjugate base form? Explain your reasoning.

b. In beaker X the oil layer is yellow, whereas in beaker Y, the oil layer is colorless. Explain these observations in terms of both acid-base equilibria and interparticle forces.

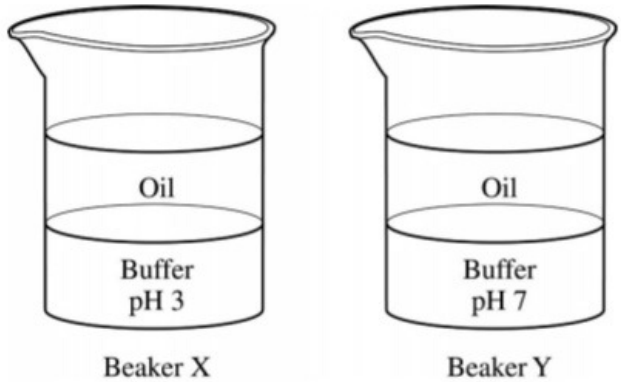
frq9

18.

$$HIn(aq) + H_2O(l) \rightleftharpoons In^-(aq) + H_3O^+(aq)$$

yellow blue

The indicator HIn is a weak acid with a pK_a value of 5.0. It reacts with water as represented in the equation above. Consider the two beakers below. Each beaker has a layer of colorless oil (a nonpolar solvent) on top of a layer of aqueous buffer solution. In beaker X the pH of the buffer solution is 3, and in beaker Y the pH of the buffer solution is 7. A small amount of HIn is placed in both beakers. The mixtures are stirred well, and the oil and water layers are allowed to separate.



a. What is the predominant form of HIn in the aqueous buffer in beaker Y, the acid form or the conjugate base form? Explain your reasoning.

b. In beaker X the oil layer is yellow, whereas in beaker Y the oil layer is colorless. Explain these observations in terms of both acid-base equilibria and interparticle forces.

19.

Compound	Melting Point (°C)
LiI	449
KI	686
LiF	845
NaF	993

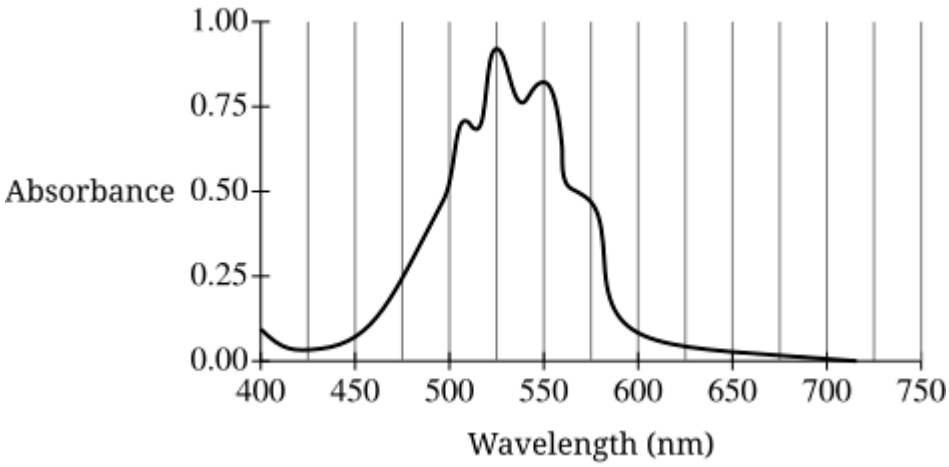
A student learns that ionic compounds have significant covalent character when a cation has a polarizing effect on a large anion. As a result, the student hypothesizes that salts composed of small cations and large anions should have relatively low melting points.

a. Select two compounds from the table and explain how the data support the student's hypothesis.

b. Identify a compound from the table that can be dissolved in water to produce a basic solution. Write the net ionic equation for the reaction that occurs to cause the solution to be basic.

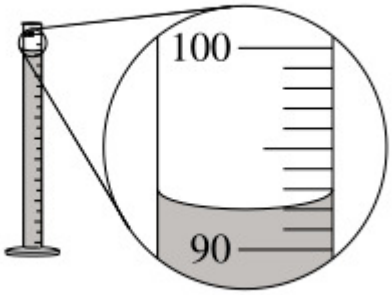
frq9

20. A student wants to determine the concentration of permanganate, $MnO_4^-(aq)$, in a solution. The student plans to use colorimetric analysis because solutions containing $MnO_4^-(aq)$ have a purple color.



(a) To determine the optimum wavelength for an experiment that measures the concentration of $MnO_4^-(aq)$, the student takes a sample of the solution and measures the amount of light absorbed by the sample over a range of wavelengths. The data are plotted in the graph shown. Identify the optimum wavelength that the student should use for the experimental procedure.

(b) The student uses a stock solution of $2.40 \times 10^{-3}\ M\ KMnO_4(aq)$ to prepare the standard solutions of $MnO_4^-(aq)$ that are needed to construct a calibration curve.



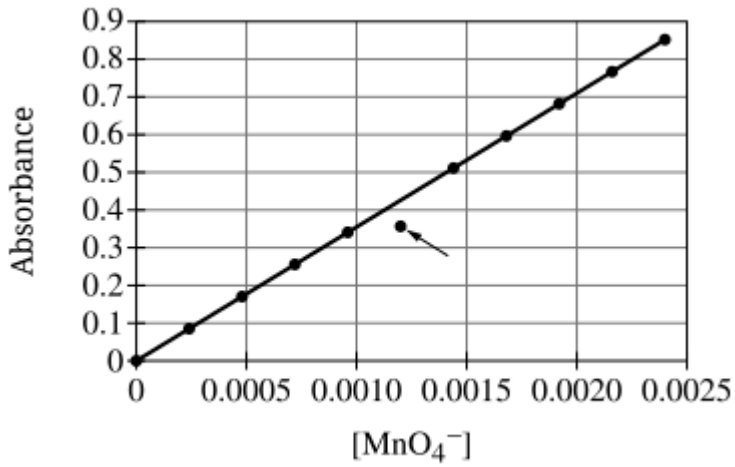
(i) The student uses a $100.0\ mL$ graduated cylinder to measure a certain volume of $KMnO_4(aq)$ stock solution, as shown in the diagram given. What volume should the student record?
(ii) Calculate the volume, in mL, of $2.40 \times 10^{-3}\ M\ KMnO_4(aq)$ that is required to produce $100.0\ mL$ of a standard $1.68 \times 10^{-3}\ M\ MnO_4^-(aq)$ solution.

frq9

The student designs the following procedure to produce a calibration curve.

Step 1: Prepare several standard solutions that have known $MnO_4^-(aq)$ concentrations by dilution of the stock solution.
Step 2: Rinse the cuvette with distilled water.
Step 3: Rinse the cuvette with the standard solution and fill the cuvette with the standard solution.
Step 4: Measure the absorbance of the standard solution with the colorimeter.
Step 5: Repeat steps 2-4 for each of the standard solutions.

The data are plotted in the calibration curve shown. One of the data points (indicated with an arrow) on the calibration curve is below the line of best fit.



(c) Assuming that all lab equipment is functioning properly, identify which one of the procedural steps the student could have executed incorrectly that would explain why the marked data point is below the line of best fit. Justify your answer.

frq9

21. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

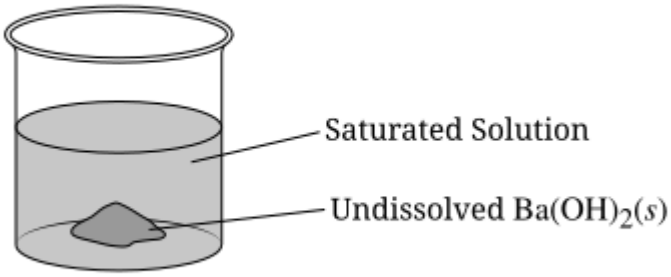
To determine the value of the solubility product constant, K_{sp}, a student prepares a saturated solution by dissolving $\mathrm{Ba(OH)_2}(s)$ in water. The K_{sp} expression is $K_{sp} = \left[\mathrm{Ba^{2+}}\right]\left[\mathrm{OH^-}\right]^2$.

(a) Write the balanced equation for the dissociation reaction that is consistent with the K_{sp} expression.

(b) The student determines that the solubility of $\mathrm{Ba(OH)_2}(s)$ is $18.5 \quad \mathrm{g/L}$ at $25^\circ\mathrm{C}$.

(i) Calculate the molar solubility of $\mathrm{Ba(OH)_2}(s)$ in $\mathrm{mol/L}$ at $25^\circ\mathrm{C}$.

(ii) Calculate the value of K_{sp} for $\mathrm{Ba(OH)_2}(s)$ at $25^\circ\mathrm{C}$.

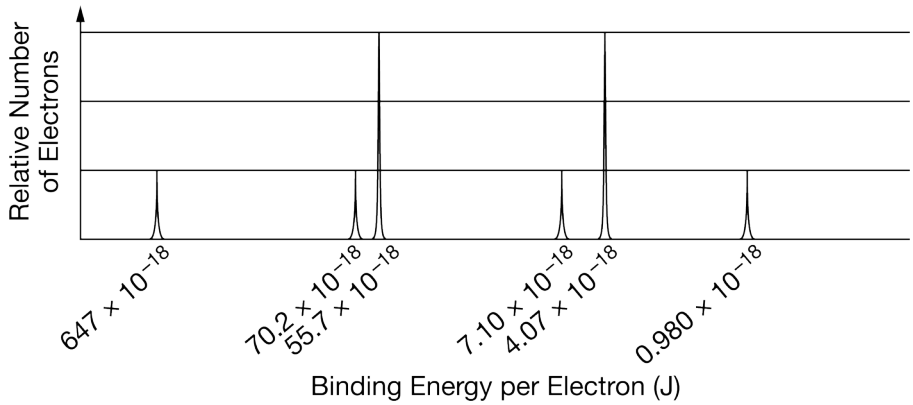


(c) The student creates a saturated solution of $\mathrm{Ba(OH)_2}$ in water, as shown in the diagram. The student claims that the amount of undissolved $\mathrm{Ba(OH)_2}(s)$ will decrease when $\mathrm{HNO_3}(aq)$ is added. Do you agree or disagree with the student's claim? Justify your answer with a comparison of Q and K_{sp}.

frq9

22. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

The complete photoelectron spectrum of an element in its ground state is represented below.



(a) Based on the spectrum,

(i) write the ground-state electron configuration of the element, and

(ii) identify the element.

(b) Calculate the wavelength, in meters, of electromagnetic radiation needed to remove an electron from the valence shell of an atom of the element.

frq9

23. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student is interested in studying the $Cu^{2+}(aq)$ ion, which is important in the environment.

$Cu(NO_3)_2(aq) + K_3PO_4(aq) \rightarrow$

(a) First, the student carries out a precipitation reaction. A solution of copper(II) nitrate is mixed with a solution of potassium phosphate, as represented above, and a blue precipitate forms. Write the balanced net ionic equation for the process.

(b) In a separate experiment the student decides to precipitate the $Cu^{2+}(aq)$ as $Cu(OH)_2(s)$. To make sure that $Cu(OH)_2(s)$ will precipitate, the student looks up the K_{sp} value for $Cu(OH)_2(s)$, as shown below.

$Cu(OH)_2(s) \rightleftarrows Cu^{2+}(aq) + 2\ OH^-(aq) \qquad K_{sp} = 2.2 \times 10^{-20}$

(i) Write the K_{sp} expression for the dissolving of $Cu(OH)_2(s)$.

(ii) Calculate the concentration of $Cu^{2+}(aq)$ in a saturated $Cu(OH)_2$ solution in mol/L.

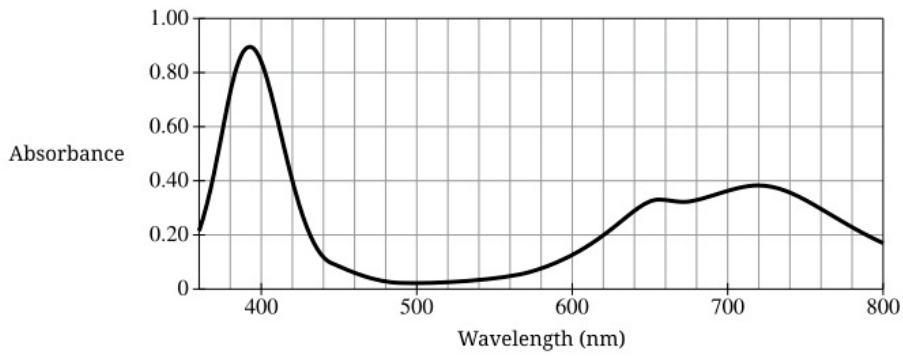
(c) The student incorrectly proposes that lowering the pH of the saturated $Cu(OH)_2$ solution will increase the amount of precipitate that forms. Explain why the student is incorrect.

frq9

24. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

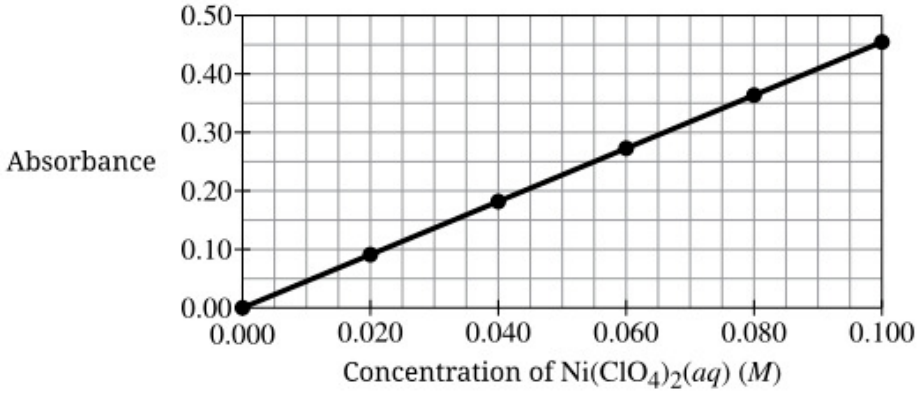
For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student performs a spectrophotometry experiment using solutions of $\mathrm{Ni}\,(\mathrm{ClO_4})_2(aq)$. The student needs to make a standard curve and determine the concentration of $\mathrm{Ni}\,(\mathrm{ClO_4})_2(aq)$ in a sample with an unknown concentration.



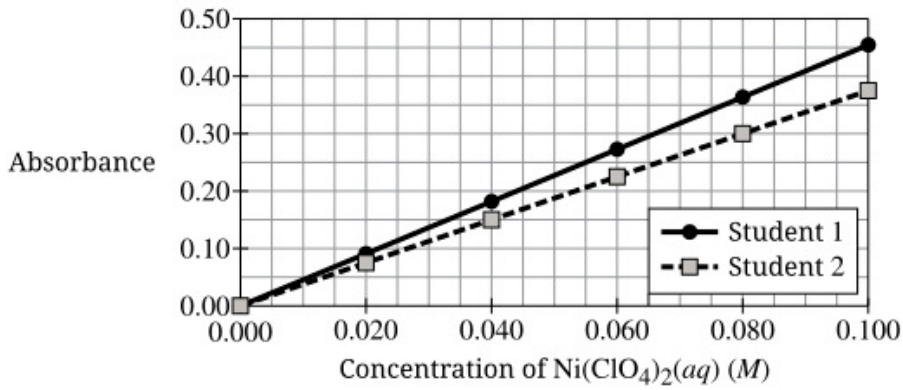
(a) Based on the absorbance spectrum of $\mathrm{Ni}\,(\mathrm{ClO_4})_2(aq)$ shown in the preceding graph, what wavelength should the student use in the experiment to achieve the greatest precision?

(b) To make the standard curve, the student must prepare $\mathrm{Ni}\,(\mathrm{ClO_4})_2(aq)$ solutions with different concentrations. The student has a standard solution of $\mathrm{Ni}\,(\mathrm{ClO_4})_2(aq)$ with a concentration of $0.198\ M$. Calculate the volume, in mL, of $0.198\ M\quad \mathrm{Ni}\,(\mathrm{ClO_4})_2(aq)$ that is needed to prepare $250.0\ \mathrm{mL}$ of $0.0400\ M\quad \mathrm{Ni}\,(\mathrm{ClO_4})_2(aq)$.



frq9

(c) The standard curve generated by the student is provided above. The student measures an absorbance of 0.30 for the unknown solution. Using the standard curve provided, determine the molarity of $\mathrm{Ni}\left(\mathrm{ClO}_4\right)_2(aq)$ in the unknown solution.



(d) A second student repeats the procedure using the same spectrophotometer but makes an error, which results in the dashed line shown in the graph above. Assuming that the first student's data (shown by the solid line) are correct, identify one error that could account for the second student's standard curve being less steep than that obtained by the first student.

frq9

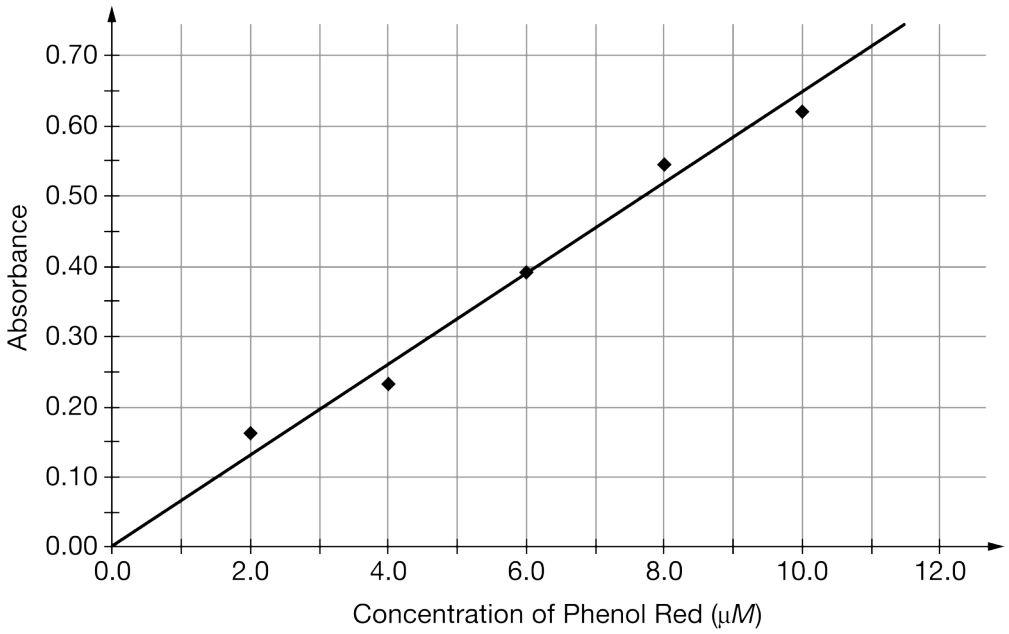
25. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student is given a solution of phenol red of unknown concentration. Solutions of phenol red are bright pink under basic conditions.

(a) The student uses a pH probe and determines that the pH of the solution is 11.20. Calculate the hydrogen ion concentration, $[\mathrm{H}^+]$, in the solution of phenol red.

The student analyzes a sample of the solution using a spectrophotometer set at a wavelength of $559\ \mathrm{nm}$, the wavelength of maximum absorbance for phenol red. The measured absorbance of the phenol red solution at $559\ \mathrm{nm}$ is 0.35.

(b) Based on the calibration curve shown below, what is the concentration of the solution in micromoles per liter (μM) ?



(c) If the student mixed $10\ \mathrm{mL}$ of distilled water with $10\ \mathrm{mL}$ of the sample, would this diluted solution have an absorbance greater than, less than, or equal to the absorbance of the original solution? Justify your answer.

(d) If the student measured the absorbance of the solution at a wavelength of $650\ \mathrm{nm}$, would the absorbance be greater than, less than, or equal to the absorbance of the solution at $559\ \mathrm{nm}$? Justify your answer.